Budget-split deep ensembles outperform snapshot ensembles and weight averaging for calibrated uncertainty*

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Abstract

Practitioners who need calibrated predictive uncertainty usually face a fixed training budget, yet method comparisons typically let every method train until it stops improving. We ask how a fixed budget should be spent. Under an identical budget of 60 SGD epochs we compared four allocation strategies for a multilayer perceptron on Fashion-MNIST: one long-trained network (Single); three independently restarted 20-epoch networks combined by prediction averaging (DeepEns); three cycle-end snapshots of one cyclic-learning-rate run (Snapshot); and the uniform weight average of those same snapshots (SWA). Across ten seeded replications, every strategy improved test negative log-likelihood (NLL) over Single (Holm-adjusted exact permutation $P = 0.0098$), with the ordering DeepEns > Snapshot > SWA > Single on every in-distribution metric. The ordering was robust across budgets (30, 60, 120 epochs), ensemble sizes ($k = 3$, 5) and a reversed dataset direction (MNIST in-distribution), although on the near-saturated MNIST task only DeepEns separated from Single (ΔNLL $= -0.016$, adjusted $P = 0.0098$). Only prediction averaging improved out-of-distribution (OOD) detection, on both OOD probes and in both directions; SWA showed no reliable improvement. Post-hoc temperature scaling fitted per condition on validation data repaired much of the in-distribution miscalibration of every condition yet left the budget-split ensemble significantly best (NLL 0.289 versus 0.333 for calibrated Single, adjusted $P = 0.0098$). The evidence is bounded to small networks and grayscale image tasks, but within that regime it supports a simple rule: split a fixed budget across independent restarts rather than extending training or averaging weights.

Keywords: Deep ensembles; Snapshot ensembles; Stochastic weight averaging; Uncertainty calibration; Budget-matched evaluation; Out-of-distribution detection

1 Introduction

Modern classifiers achieve high accuracy yet assign miscalibrated confidence: networks trained for many epochs with small mini-batches become systematically overconfident, so their predicted probabilities cannot be interpreted at face value (Guo et al., 2017; Abdar et al., 2021). Because downstream decisions, selective prediction and abstention all consume these probabilities, miscalibration is a deployment risk rather than a cosmetic defect, and a large body of work now addresses uncertainty quantification in deep learning (Abdar et al., 2021; Ovadia et al., 2019). Among the available remedies, three families dominate empirical practice. Deep ensembles average the predictions of independently trained networks and set a strong but compute-expensive baseline (Lakshminarayanan et al., 2017; Fort et al., 2019). Single-trajectory approximations reuse one training run: snapshot ensembles collect the checkpoints left behind by cyclic learning-rate schedules (Huang et al., 2017; Loshchilov and Hutter, 2017), and stochastic weight averaging (SWA) averages network weights

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along a (typically cyclic) trajectory (Izmailov et al., 2018; Wilson and Izmailov, 2020; Maddox et al., 2019). Variational and Bayesian approximations such as Monte Carlo dropout form a third family (Gal and Ghahramani, 2016; Welling and Teh, 2011).

These families are usually compared at their own best operating points: each method is tuned until it stops improving, and the reported trade-off is accuracy or calibration against the *added* cost of the method (Ashukha et al., 2020). Practitioners, however, face the inverse problem: a fixed compute allocation and a choice of how to spend it. Under a budget of B epochs one may train one network for B epochs, k networks for B/k epochs each, one cyclic run of B epochs whose cycle-end snapshots form an ensemble, or the weight average of those snapshots. Benchmark studies have mapped the performance of uncertainty methods under distribution shift and documented the pitfalls of in-domain evaluation (Ovadia et al., 2019; Ashukha et al., 2020), but they vary budgets across methods. The question of which *allocation* of a fixed budget buys the best-calibrated uncertainty is therefore only partially answered, and the mechanism behind the expected ordering—member quality versus member diversity, prediction averaging versus weight averaging—is rarely isolated within one controlled experiment.

We addressed this gap with a four-way, budget-matched, seed-paired comparison in which every strategy consumed exactly B epochs of the same optimizer, architecture, data split and learning-rate ceiling. Two design choices make the contrasts unusually clean. First, the first ensemble member of the budget-split ensemble and the first cyclic snapshot are the same network (verified bit-identical), so the DeepEns–Snapshot contrast isolates the effect of restarting training independently versus continuing one trajectory. Second, Snapshot and SWA combine the *identical* checkpoints, so their contrast isolates the combination mechanism—prediction space versus parameter space—with no trajectory noise. The pre-registered core setting (Fashion-MNIST, $B = 60$, $k = 3$, ten seeds) is extended here along four axes: total budget ($B \in \{30, 60, 120\}$), ensemble size ($k \in \{3, 5\}$), dataset direction (MNIST in-distribution with Fashion-MNIST as the shift probe, in addition to the reverse), OOD probe identity (MNIST and KMNIST), and a post-hoc temperature-scaling baseline fitted per condition on validation data (Guo et al., 2017; Zhang et al., 2020).

The contributions of this paper are ascriptive rather than algorithmic; we offer a controlled measurement and a mechanism account, not a new method.

- **A compute-exact four-way comparison design** in which the budget-split ensemble, the snapshot ensemble and the weight average share training compute, data and (in part) checkpoints, with seed-paired exact permutation inference throughout (Sect. 3).
- **A robust ordering result.** Splitting a fixed budget across independent restarts produced the best-calibrated uncertainty on every in-distribution metric, and this ordering was stable across budgets, ensemble sizes and both dataset directions tested (Sect. 4.1, 4.2).
- **A boundary condition.** On the near-saturated MNIST task the single-trajectory strategies stopped separating from the single long-trained network, whereas the budget-split ensemble remained significantly better; the ensemble effect in this regime requires headroom for longer training to damage calibration (Sect. 4.2).
- **A separation of combination mechanisms under shift.** Averaging predictions improved OOD detection on every probe–direction combination, while averaging the same weights showed no reliable improvement (Sect. 4.4).
- **A post-hoc recalibration stress test.** Temperature scaling fitted per condition compressed in-distribution differences but did not close them, and left OOD detection essentially unchanged (Sect. 4.3).

The paper proceeds as follows. Section 2 positions the study against prior work. Section 3 specifies the design, metrics and statistical analysis. Section 4 reports the results organized by research question, Sect. 5 discusses interpretation, limitations and practical guidance, and Sect. 6 concludes.

2 Related work

Deep ensembles and their mechanism Lakshminarayanan et al. (2017) showed that averaging the predictive distributions of independently trained networks is a simple, scalable and strongly competitive uncertainty estimator, and subsequent studies confirmed its robustness under dataset shift (Ovadia et al., 2019). Fort et al. (2019) analysed *why* restarts help: independently initialised networks converge to different basins of the loss landscape, so their errors decorrelate. That analysis, however, treats ensembles as an *added* cost; the present study instead asks what restarts buy when the budget is fixed and each member is correspondingly shorter. Ashukha et al. (2020) is closest in spirit: their "pitfalls" study of in-domain uncertainty includes compute-matched single-versus-ensemble comparisons and finds ensembles remain preferable. Our design differs in two respects: we include the two single-trajectory strategies (snapshot ensembles and SWA) that a budget-constrained practitioner would actually choose between, and we hold checkpoints identical between the prediction-averaging and weight-averaging conditions so that the combination mechanism is isolated exactly.

Single-trajectory ensembles Snapshot ensembles exploit warm restarts (Loshchilov and Hutter, 2017): one cyclic-learning-rate run yields several "free" models at the cycle minima (Huang et al., 2017). SWA instead averages the weights of checkpoints along a trajectory and lands in flatter, wider regions associated with better generalisation (Izmailov et al., 2018); Bayesian readings and Gaussian generalisations of the same idea include the work of Wilson and Izmailov (2020) and Maddox et al. (2019). Related approximations reduce ensemble cost by sharing parameters, e.g. BatchEnsemble (Wen et al., 2020). These works establish what each strategy achieves at its own budget; the budget-matched trade-off among them, and in particular the contrast between combining the same checkpoints in function space versus parameter space, is what we measure. We also evaluate a denser SWA variant that averages every epoch-end checkpoint, following standard SWA practice (Izmailov et al., 2018), as an exploratory bridge.

Calibration and post-hoc recalibration Overconfidence of modern networks and the expected calibration error were popularised as practical metrics by Guo et al. (2017), who also introduced the temperature-scaling (TS) post-hoc fix we adopt: a single temperature fitted on held-out data to minimise validation NLL. Naeini et al. (2015) provide the binning foundation for calibration measurement. Compositional post-hoc calibration (Zhang et al., 2020) shows that calibration and ensembling interact; the question we isolate is narrower and practical: does per-condition TS remove the advantage that budget-split restarts hold over longer training, or does it merely compress the differences? Our setting extends the classic observation that ensembles remain better calibrated than recalibrated single models (Lakshminarayanan et al., 2017) to the budget-matched regime.

Uncertainty under distribution shift The maximum-softmax score of Hendrycks and Gimpel (2017) is the canonical baseline for detecting out-of-distribution inputs, and confidence-based scores are known to be brittle (Nguyen et al., 2015). Ovadia et al. (2019) benchmarked a broad set of uncertainty methods under shift, finding deep ensembles the most robust; that study, however, varies training budgets across methods and does not include the checkpoint-matched snapshot-versus-SWA contrast that separates prediction averaging from weight averaging here. Our OOD analysis uses two probes in each dataset direction (MNIST and KMNIST against Fashion-MNIST in-distribution, and the reverse), which lets us check that conclusions are not specific to one probe.

Positioning No prior study that we are aware of isolates, within a single controlled experiment, (i) the allocation of a fixed budget across restarts versus one long run, (ii) the combination of identical checkpoints by prediction averaging versus weight averaging, and (iii) the interaction of both with post-hoc recalibration and OOD detection. We claim novelty for the *comparison design and measurement*, not for any of the component methods, all of which are standard.

3 Study design and methods

3.1 Budget-matched conditions

The total training budget B is fixed, and four allocation strategies consume it in different ways (Fig. 1a). SINGLE trains one network for B epochs under a single cosine anneal from learning rate 0.1 to 0. DEEPENS(budget-split deep ensemble) trains k independently initialised networks for B/k epochs each, each under its own complete cosine anneal, and averages their predictive distributions. SNAPSHOTtrains one network for B epochs under k cosine cycles (Loshchilov and Hutter, 2017) and keeps the k cycle-end checkpoints (Huang et al., 2017). SWAtakes the uniform weight average of those same k checkpoints (Izmailov et al., 2018). Throughout, the pre-registered core setting is Fashion-MNIST with $B = 60$ and $k = 3$; the extension settings vary $B \in \{30, 120\}$, $k = 5$, and the in-distribution dataset (Sect. 3.2).

Two design details strengthen the comparisons. First, the first cosine cycle of the cyclic run is, by construction, a complete B/k-epoch anneal with a specified initialisation and data order; we verified programmatically that this checkpoint is bit-identical to an independent B/k-epoch run under the same seed, and used it as both the first SNAPSHOT member and the first DEEPENS member. The implementation is therefore compute-exact ($2B + (k-1)B/k$ epochs per replicate) while the two ensemble conditions differ, from the second member onward, precisely in whether subsequent members restart independently or continue the trajectory. Second, SNAPSHOT and SWA are derived from the same run, so their contrast measures only how the identical checkpoints are combined: by averaging predictions (function space) or weights (parameter space). We note that our SWA averages exactly k cycle-end checkpoints in order to remain checkpoint-matched with SNAPSHOT; this is a restricted SWA schedule relative to the many-checkpoint averages of standard practice (Izmailov et al., 2018), and an exploratory dense variant averaging every epoch-end checkpoint is reported in the supplementary material.

3.2 Datasets and splits

Two dataset directions are used. In the primary direction, the in-distribution task is Fashion-MNIST (Xiao et al., 2017) (60,000 training and 10,000 test images, 28×28 grayscale, ten classes) and the OOD probes are the MNIST (LeCun et al., 1998) and KMNIST (Clanuwat et al., 2018) test sets (10,000 images each). In the reversed direction, MNIST is the in-distribution task and the Fashion-MNIST and KMNIST test sets serve as probes. KMNIST (handwritten cursive Japanese characters) provides a second, visually closer shift probe whose label spaces do not overlap with either in-distribution task. In each direction, the 60,000-image training set was split once, with a fixed documented split seed, into 54,000 training and 6,000 validation images. Images were scaled to $[0, 1]$ and standardised with the mean and standard deviation of the corresponding training split; OOD probes were standardised with the *in-distribution* statistics, matching the deployment situation in which a shift is not known in advance. The validation split was used only for monitoring, learning curves and temperature fitting (Sect. 3.4); no early stopping, hyperparameter selection or model selection was performed on validation or test data. The test set and the OOD probes were evaluated exactly once per condition per replicate, after all training was complete.

3.3 Architecture and training

All networks were multilayer perceptrons with layers 784–128–64–10 and ReLU activations (≈110k parameters), without dropout or batch normalisation. Training used mini-batch SGD with batch size 256 (211 steps per epoch on 54,000 images), momentum 0.9 and weight decay 10^{-4} (coupled). Learning rates followed per-step cosine schedules from $\eta_{\max} = 0.1$ to 0: a single B-epoch anneal for SINGLE, k cycles for the cyclic run, and one B/k-epoch anneal per DEEPENS member. Initialisation used PyTorch's default Kaiming-uniform scheme; the data order of each run was generated from a dedicated seeded generator. Hyperparameters were fixed from common practice before any test-set

evaluation, and a pilot confirmed training stability using training and validation losses only. On two CPU cores, a $B = 60$ replicate took 72–87 s of wall-clock time; the extension settings required about 9,000 training epochs (69 minutes of wall-clock time), and together with the pre-registered core and the supplementary dense-SWA pass the complete study comprises roughly 11,200 epochs, under two hours of single-machine CPU time.

3.4 Evaluation metrics

All metrics were computed on the full test set (and, for OOD, the full probe) with probabilities evaluated in float64. NLL is the mean negative log-probability of the true class under the (ensemble-mean) predictive distribution. Accuracy is the fraction of argmax-correct predictions. The Brier score is the mean over samples of the summed squared error between the predictive distribution and the one-hot label. Expected calibration error (ECE) uses 15 equal-width bins over maximum predicted probability (Naeini et al., 2015; Guo et al., 2017), with 10 bins as a sensitivity check. OOD detection uses the AUROC of the maximum-softmax score (Hendrycks and Gimpel, 2017) with in-distribution samples as the positive class, and the negative-entropy score as a sensitivity check. Ensemble disagreement is the mean over test samples and member pairs of the Jensen–Shannon divergence between member predictive distributions (natural logarithm). Ensemble predictive distributions are the arithmetic mean of member softmax outputs.

Temperature scaling Following Guo et al. (2017), we fit a single temperature T per condition and replicate by minimising NLL on the 6,000 validation images, and apply it to the test and probe sets. The scaled predictive distribution is $\mathrm{softmax}(z/T)$ where z is the effective logit vector of the condition: the raw logits for single-model conditions, and the logarithm of the averaged probabilities for ensemble conditions (an averaged ensemble has no raw logits of its own). For single networks this coincides exactly with standard temperature scaling, because the softmax is shift-invariant and $\log p$ differs from the raw logits only by a constant per sample. The fitted temperature itself is reported as an overconfidence diagnostic. This per-condition TS is the strongest simple post-hoc recalibration baseline (Guo et al., 2017; Zhang et al., 2020): it asks whether the between-strategy differences survive when every strategy is allowed its own best recalibration.

3.5 Statistical analysis

The analysis of the core setting (including the contrast set, the primary outcome and the multiplicity strategy) was written and stored before any test-set evaluation; the only amendment (increasing replications from five to ten) was made after a timing pilot, before any outcome was observed. The independent experimental unit is the seed; each seed contributes one value per condition per metric, and all contrasts are within-seed paired differences. We report mean differences with two-sided 95% t-intervals ($n-1 = 9$ degrees of freedom), exact two-sided sign-flip permutation P values enumerating all 2^{10} sign patterns, and Holm–Bonferroni adjustment across the five pre-specified contrasts applied to the primary outcome (NLL). The five contrasts are DeepEns−Single, Snapshot−Single, SWA−Single, DeepEns−Snapshot and Snapshot−SWA; the first three measure what each strategy buys over spending the whole budget on one network, and the last two locate the ordering. The extension settings (B, k, dataset direction, probes) and the temperature-scaled analyses were not part of the pre-registered plan and are labelled as such; each extension setting is analysed with the same machinery and its own Holm family over the same five contrasts on NLL, and the TS family on TS-NLL is likewise Holm-corrected. Sensitivity analyses (10-bin ECE, entropy-based AUROC score, dense SWA) were specified in advance or, for dense SWA, labelled exploratory. No data were excluded; no condition, replicate or sample was dropped at any stage.

Table 1: **Test-set performance by condition in the core setting** (mean ± s.d. over $n = 10$ replicates; 10,000 Fashion-MNIST test images; OOD probes: 10,000 MNIST and 10,000 KMNIST images). JSD and member NLL are defined only for the two ensemble conditions.

Metric	Single	DeepEns	Snapshot	SWA
Test NLL (nats)	0.5265 ± 0.0159	$\mathbf{0.3020 \pm 0.0019}$	0.3485 ± 0.0067	0.3868 ± 0.0085
Accuracy	0.8895 ± 0.0015	$\mathbf{0.8965 \pm 0.0012}$	0.8935 ± 0.0015	0.8926 ± 0.0018
Brier score	0.1796 ± 0.0020	$\mathbf{0.1504 \pm 0.0011}$	0.1596 ± 0.0024	0.1646 ± 0.0028
ECE (15 bins)	0.0702 ± 0.0021	$\mathbf{0.0236 \pm 0.0008}$	0.0444 ± 0.0025	0.0517 ± 0.0025
AUROC (MSP), MNIST probe	0.7533 ± 0.0172	$\mathbf{0.8502 \pm 0.0049}$	0.7956 ± 0.0139	0.7504 ± 0.0168
AUROC (MSP), KMNIST probe	0.6951 ± 0.0184	$\mathbf{0.8048 \pm 0.0067}$	0.7536 ± 0.0131	0.7063 ± 0.0132
AUROC (neg. entropy), MNIST	0.7562 ± 0.0172	$\mathbf{0.8657 \pm 0.0055}$	0.8059 ± 0.0144	0.7573 ± 0.0171
Member mean NLL	n/a	0.3379 ± 0.0022	0.4039 ± 0.0087	n/a
Disagreement JSD (nats)	n/a	0.0150 ± 0.0003	0.0106 ± 0.0003	n/a

3.6 Reproducibility

Experiments used PyTorch 2.14.0 (CPU), NumPy 2.1.3, SciPy 1.14.1 and scikit-learn 1.5.2 on Python 3.12 (two CPU cores, no GPU). Every training run is deterministic given its replicate seed; the v2 pipeline used for the extension settings reproduces the pre-registered v1 records exactly (a seed-0 re-run matched all 28 recorded metrics bit-for-bit, and the shared first member passed bit-identity checks in both pipelines). Training, analysis and figure scripts, the raw per-replicate metric records (JSONL), the aggregated analysis outputs and the per-panel figure source data are released in a public repository (`https://github.com/43aquarius/IJMLC`); datasets are publicly available (Xiao et al., 2017; LeCun et al., 1998; Clanuwat et al., 2018).

4 Results

4.1 Budget-matched comparison in the core setting

All three ensembling strategies improved the primary outcome, test NLL, over Single at matched budget (Table 2, Fig. 1b): mean paired differences were -0.2245 (95% CI $[-0.2360, -0.2130]$) for DeepEns, -0.1780 for Snapshot and -0.1396 for SWA, each with exact permutation $P = 0.0020$ and Holm-adjusted $P = 0.0098$ across the five pre-specified contrasts. The strategies were also strictly ordered: DeepEns improved on Snapshot by -0.0465 ($[-0.0505, -0.0425]$) and Snapshot improved on SWA by -0.0384 ($[-0.0407, -0.0360]$), both with adjusted $P = 0.0098$. The single long-trained network combined the highest per-epoch accuracy trajectory with the worst predictive log-likelihood: its NLL of 0.5265 ± 0.0159 (mean ± s.d.) coexisted with accuracy of 0.8895 ± 0.0015, reflecting severe overconfidence rather than classification error.

The same ordering held on every secondary in-distribution metric (Table 1). Accuracy improved modestly but consistently, the Brier score and ECE followed the NLL ordering, and the effect sizes were stable across replicates (all ten replicates favoured every strategy over Single for NLL). The reliability diagrams locate the differences in the confidence scale (Fig. 1c, Supplementary Fig. S1): the single network's predictions clustered at high confidence with accuracy below that confidence (ECE 0.0702 ± 0.0021; a typical predicted confidence of about 0.96 was matched by roughly 0.89 empirical accuracy), DeepEns nearly restored calibration (ECE 0.0236 ± 0.0008, a 66% reduction), Snapshot removed about a third of the error and SWA about a quarter. These calibration gains, rather than accuracy, carried most of the NLL improvement.

4.2 Robustness across budgets, ensemble sizes and dataset direction

The extension settings test whether the core ordering is an artefact of one budget, one ensemble size or one dataset direction (Fig. 2, Table 3). Across budgets $B \in \{30, 60, 120\}$ the NLL ordering

Table 2: **Pre-specified contrasts on the primary outcome (test NLL) in the core setting.** Mean paired difference Δ (condition A − condition B), two-sided 95% confidence interval, exact sign-flip permutation P (all 2^{10} sign patterns), and Holm-adjusted P across the five contrasts ($n = 10$ replicates).

Contrast (A − B)	Δ NLL	95% CI	P_{perm}	P_{Holm}
DeepEns − Single	−0.2245	[−0.2360, −0.2130]	0.0020	0.0098
Snapshot − Single	−0.1780	[−0.1902, −0.1658]	0.0020	0.0098
SWA − Single	−0.1396	[−0.1514, −0.1279]	0.0020	0.0098
DeepEns − Snapshot	−0.0465	[−0.0505, −0.0425]	0.0020	0.0098
Snapshot − SWA	−0.0384	[−0.0407, −0.0360]	0.0020	0.0098

DeepEns > Snapshot > SWA > Single was preserved, with all five contrasts Holm-adjusted $P = 0.0098$ at every budget. The budget-split advantage *grew* with the budget: the DeepEns−Single difference widened from −0.0754 at $B = 30$ to −0.2245 at $B = 60$ and −0.3317 at $B = 120$, because longer training made the single network progressively more overconfident (its NLL rose from 0.3795 to 0.5265 to 0.6672 while its accuracy was flat) while the 20- or 40-epoch restart members stayed well calibrated. Enlarging the ensemble from $k = 3$ to $k = 5$ at fixed budget improved every strategy (Fig. 2b) and preserved the ordering (DeepEns −0.2290 vs. Single, adjusted $P = 0.0098$).

Reversing the dataset direction exposed a boundary condition. With MNIST as the in-distribution task (accuracy 0.982 for Single; ECE 0.0098), only DeepEns separated from Single on NLL (−0.0158, 95% CI [−0.0183, −0.0134], adjusted $P = 0.0098$); the single-trajectory strategies showed no reliable improvement (Snapshot −0.0015, $P = 0.41$; SWA +0.0017, $P = 0.41$; Fig. 4a). The ensemble effect that longer training damages calibration and averaging repairs it—requires headroom that the near-saturated MNIST task does not provide: the single 60-epoch network was already well calibrated there (ECE 0.0098 versus 0.0702 on Fashion-MNIST). The budget-split restarts nevertheless remained the best allocation even on MNIST: DeepEns still beat Snapshot (−0.0143, adjusted $P = 0.0098$), and by adjacency also SWA (the Snapshot−SWA contrast was itself significant at adjusted $P = 0.0098$).

4.3 Post-hoc temperature scaling compresses but does not close the gaps

A natural objection is that the between-strategy differences in NLL reflect fixable overconfidence rather than information content: a single temperature fitted on validation data repairs much of a network's miscalibration at negligible cost (Guo et al., 2017). We therefore fitted one temperature per condition and replicate (Fig. 3) and re-evaluated all metrics. Temperature scaling behaved as expected: it helped every condition, and most dramatically the single network, whose NLL fell from 0.5265 to 0.3325 and whose ECE fell from 0.0702 to 0.0111. The fitted temperatures mirror the overconfidence ordering (Single 2.46, SWA 1.79, Snapshot 1.58, DeepEns 1.25): the strategies that looked worst calibrated uncalibrated needed the most cooling.

Two findings survive the recalibration, however. First, the ordering persisted: after per-condition TS, DeepEns still attained the lowest NLL (0.2891 ± 0.0020) and every contrast in the five-contrast family remained significant at Holm-adjusted $P = 0.0098$ (DeepEns−Single −0.0434, 95% CI [−0.0468, −0.0400]; Fig. 3a). Budget-split restarts therefore do not merely produce a conveniently recalibratable distribution; averaging independent restarts yields information that a monotone rescaling of any single trajectory cannot recover. Second, recalibration bought nothing where the deployment risk lives: TS changed OOD AUROC by at most +0.012 on any condition or probe (e.g. Single 0.7533 → 0.7620 on the MNIST probe, DeepEns 0.8502 → 0.8566), leaving the between-strategy OOD differences intact (Sect. 4.4). The same two conclusions held in every extension setting: after TS, DeepEns remained significantly better than Single on NLL at $B = 30$ (−0.0114), $B = 120$ (−0.0567), $k = 5$ (−0.0384) and in the MNIST direction (−0.0071), all adjusted $P = 0.0098$. In the low-budget setting the TS-adjusted strategies were close together

Table 3: **Extension settings: contrasts versus Single on test NLL** (mean paired difference, 95% CI, Holm-adjusted exact permutation P within each setting's five-contrast family; $n = 10$ replicates each). The core setting row repeats Table 2 for comparison. All within-setting pairwise contrasts (DeepEns−Snapshot, Snapshot−SWA) are significant at adjusted $P = 0.0098$ in every Fashion-MNIST setting and in the MNIST direction.

Setting	Contrast vs. Single	Δ NLL	95% CI	P_{Holm}
Fashion-MNIST, B=60, k=3 (core)	DeepEns	−0.2245	[−0.2360, −0.2130]	0.0098
	Snapshot	−0.1780	[−0.1902, −0.1658]	0.0098
	SWA	−0.1396	[−0.1514, −0.1279]	0.0098
Fashion-MNIST, B=30, k=3	DeepEns	−0.0754	[−0.0809, −0.0699]	0.0098
	Snapshot	−0.0671	[−0.0736, −0.0606]	0.0098
	SWA	−0.0530	[−0.0587, −0.0473]	0.0098
Fashion-MNIST, B=120, k=3	DeepEns	−0.3317	[−0.3416, −0.3219]	0.0098
	Snapshot	−0.2283	[−0.2377, −0.2188]	0.0098
	SWA	−0.1552	[−0.1658, −0.1446]	0.0098
Fashion-MNIST, B=60, k=5	DeepEns	−0.2290	[−0.2410, −0.2171]	0.0098
	Snapshot	−0.2035	[−0.2142, −0.1928]	0.0098
	SWA	−0.1673	[−0.1765, −0.1581]	0.0098
MNIST, B=60, k=3 (reversed)	DeepEns	−0.0158	[−0.0183, −0.0134]	0.0098
	Snapshot	−0.0015	[−0.0042, +0.0012]	0.41 (n.s.)
	SWA	+0.0017	[−0.0012, +0.0046]	0.41 (n.s.)

(NLL 0.299–0.313), so post-hoc calibration is a genuine partial substitute when the budget is small; at and beyond the core budget it is not.

4.4 OOD detection separates prediction averaging from weight averaging

We next asked which budget allocations buy robustness, measured as the separation between in-distribution test images and each OOD probe using the maximum-softmax score (Fig. 4b). In the core setting, DeepEns improved AUROC by +0.0969 over Single on the MNIST probe (95% CI $[+0.0847, +0.1091]$, $P = 0.0020$) and by +0.1097 ($[+0.1004, +0.1190]$) on the KMNIST probe, whereas SWA showed no reliable improvement on the MNIST probe (-0.0029, $[-0.0166, +0.0109]$, $P = 0.64$) and only a small, order-of-magnitude-smaller effect on KMNIST ($+0.0112$, $[+0.0023, +0.0202]$, $P = 0.023$ uncorrected). Snapshot occupied an intermediate position (+0.0423 and +0.0585 respectively). Because Snapshot and SWA combine the identical three checkpoints, this difference is attributable to the combination mechanism alone: averaging predictions transfers the members' ability to disagree on shifted inputs into the score, while averaging weights collapses the members into one network whose confidence behaviour under shift is essentially that of the trajectory it came from. We note that the width of the SWA interval (about 0.028 AUROC units) bounds what this design can resolve; the claim is one of absence of evidence at this precision, not of evidence of absence.

The pattern generalised across probes and directions. Across the four combinations of in-distribution task and OOD probe, DeepEns gave the best OOD detection and was the only strategy whose improvement over Single was significant on every combination (in the MNIST direction: +0.0139, $P = 0.008$, and +0.0235, $P = 0.002$); the single-trajectory strategies were statistically indistinguishable from Single on both probes of the MNIST direction ($P \geq 0.08$). The same pattern held with the negative-entropy score, so the conclusion is not specific to the score function, and the exploratory dense SWA variant (40-checkpoint average) likewise left OOD detection unchanged (Supplementary Table S2).

4.5 Mechanism: member calibration and member diversity

The design also pins down why DeepEns dominates Snapshot at matched budget (Fig. 1e,f); because the two ensembles share their first member, this contrast is carried by the later members, where the strategies differ in restarting versus continuing training. First, the individual members differed in calibration: the three independent 20-epoch restarts achieved a mean member NLL of 0.3379 ± 0.0022, whereas the second and third cyclic snapshots (taken after 40 and 60 cumulative epochs of continued training) were individually more overconfident (0.4039 ± 0.0087 averaged over members), even though their member accuracy was nearly identical (0.8912 versus 0.8916). Continued training on the same trajectory thus bought no member-level accuracy but cost member-level calibration. Second, the members were less diverse: the mean pairwise Jensen–Shannon divergence of member predictive distributions was 0.0150 ± 0.0003 nats for DeepEns but only 0.0106 ± 0.0003 nats for Snapshot, a 30% reduction in disagreement. At $k = 5$ the same two ingredients pointed the same way (member NLL 0.3212 versus 0.3796; JSD 0.0101 versus 0.0107, the diversity gap narrowing as the shorter 12-epoch members become more similar). Both quantities point in the direction that favours independent restarts; we note, however, that with $n = 10$ replicates and $k \leq 5$ members these measurements associate the mechanism with the outcome rather than establishing it causally.

5 Discussion

At a fixed training budget, this study returns a consistent allocation rule: splitting the budget across independent restarts (DeepEns) produced the most accurate and best-calibrated uncertainty in every setting we measured—both dataset directions, three budgets spanning a 4× range, and two ensemble sizes; a cyclic-schedule snapshot ensemble (Snapshot) captured most of the in-distribution benefit at one run's bookkeeping cost; and weight averaging (SWA) improved in-distribution metrics but, unlike either prediction-averaging strategy, showed no reliable improvement in OOD detection. Because the four conditions shared data, architecture, optimizer, budget and pairing by seed, and because Snapshot and SWA shared the identical checkpoints, these differences are attributable to the allocation of the budget and the mechanism of combination rather than to tuning or trajectory noise.

The mechanism measurements suggest an interpretation in terms of where the two averaging operations act. Prediction averaging operates in function space: when members disagree, the mixture flattens the predictive distribution, which is exactly the behaviour that repairs overconfidence and that produces low-confidence predictions on shifted inputs. Weight averaging operates in parameter space: it selects a flatter, lower-loss point along the trajectory (Izmailov et al., 2018; Wilson and Izmailov, 2020), which improves the network's fit–confidence trade-off in distribution, but it collapses the members' disagreement into a single confident model and thereby discards the epistemic signal that the shift detector consumes. Consistent with this reading, member-level analysis showed that continued-training snapshots are individually more overconfident than fresh restarts at equal accuracy, and less diverse from one another; the snapshot ensemble therefore starts from worse-calibrated, more correlated members than the budget-split ensemble even at the same budget.

The budget sweep adds a time dimension to this account. Longer training of the single network made it progressively more overconfident—its NLL *worsened* from 0.38 at $B = 30$ to 0.67 at $B = 120$ while accuracy was flat—so the value of redirecting that budget into restarts grew with the budget. The reversed dataset direction shows the complementary boundary: on a near-saturated task where the long-trained single network is already well calibrated, the single-trajectory strategies had nothing left to repair and stopped separating from Single, while budget-split restarts still helped. The ensemble effect documented here is therefore not universal; it is a property of regimes with calibration headroom, which include most practically deployed under-trained or heavily optimised classifiers and exclude easy saturated tasks.

Post-hoc recalibration is the strongest available objection to any calibration comparison, and our answer is two-sided. In distribution, per-condition temperature scaling is a genuine partial substitute, especially at small budgets where the TS-adjusted strategies clustered within 0.014 NLL of each other; at the core budget and above, however, the recalibrated budget-split ensemble remained significantly better than every recalibrated alternative. Under shift, temperature scaling bought nothing—it changes confidence levels, not their ranking across inputs—so the OOD advantage of prediction averaging cannot be recovered post hoc. This refines the classical observation that ensembles remain better calibrated than recalibrated single models (Lakshminarayanan et al., 2017; Guo et al., 2017; Zhang et al., 2020): in the budget-matched regime, what post-hoc calibration cannot buy is precisely the restart-induced diversity.

Our results are consistent with the literature that motivated the comparison—deep ensembles are strong but assumed expensive (Lakshminarayanan et al., 2017; Fort et al., 2019), snapshot ensembles are attractive because one run yields several models (Huang et al., 2017), and SWA improves accuracy and in-distribution calibration (Izmailov et al., 2018; Wilson and Izmailov, 2020)—but they quantify the trade-off under the constraint that matters in practice. The budget-matched framing reverses the usual cost accounting: a deep ensemble of three 20-epoch runs is not three times as expensive as one 60-epoch run; it *is* the same budget spent differently. Under that accounting, the "expensive" strategy was uniformly best in this regime, the "free" strategies captured different fractions of its benefit, and the cheapest-to-deploy single-trajectory variant (SWA) bought no shift robustness. This complements benchmark studies that vary budgets across methods (Ovadia et al., 2019; Ashukha et al., 2020) and echoes the observation that independent initialisation is the ingredient most responsible for ensemble gains (Fort et al., 2019): here that ingredient survives a three-fold reduction in per-member training.

For practitioners, the allocation rule we can support is as follows. At small scale and fixed compute, prefer independent restarts over longer training for calibrated uncertainty. If only a single trajectory is affordable, prefer its snapshot ensemble over its weight average when shift robustness matters, and apply temperature scaling regardless: it is nearly free, repairs in-distribution overconfidence, and does no harm to shift detection. Treat SWA as an accuracy-and-fit tool rather than a shift detector, consistent with evidence that OOD failures often persist across training procedures (Hendrycks and Gimpel, 2017; Nguyen et al., 2015). The rule is bounded by the scope of the evidence, which we summarise in Sect. 5.1.

5.1 Limitations

Our study used one architecture family (a small multilayer perceptron without batch normalisation or dropout); convolutional and transformer-based networks, and larger models in general, remain untested here, and the CPU-bound design is the reason. The dataset families are two grayscale 28×28 image tasks and their probes; conclusions about natural-image or cross-modal shift would require further experiments. The budget is denominated in epochs at a fixed learning-rate ceiling and optimizer family rather than wall-clock or energy; other optimizers or schedules could shift the crossover points, although not the checkpoint-matched logic of the Snapshot−SWA contrast. Ensemble sizes are $k \in \{3, 5\}$. Our SWA variant is checkpoint-matched to the snapshot ensemble by design; although a dense 40-checkpoint average did not change the qualitative conclusion, other SWA schedules might. Temperature scaling is the only post-hoc recalibration tested; more flexible calibrators (Zhang et al., 2020) could in principle compress the in-distribution gaps further. The OOD probes are dataset-level shifts between tasks of the same modality, evaluated with confidence-based scores; no corruptions, adversarial perturbations or likelihood-based methods were considered. Finally, the mechanism analysis is associational by design; a causal decomposition of the ensemble gain into member-quality and diversity components would require interventions on each ingredient separately.

6 Conclusion

We compared four ways of spending a fixed training budget on the uncertainty of one classifier, holding compute, data, optimizer and (in part) checkpoints identical across conditions. Budget-split deep ensembles gave the best-calibrated in-distribution uncertainty in every setting tested, an advantage that grew with the budget and survived per-condition temperature scaling; single-trajectory strategies captured part of the benefit on tasks with calibration headroom and none of it on a saturated task; and only prediction averaging improved out-of-distribution detection, on every probe and direction tested, while weight averaging of the same checkpoints showed no reliable improvement. The paired, seed-replicated, checkpoint-matched template used here is directly reusable for larger networks, harder shifts and richer recalibration families, and we release all code and raw records to make such extensions cheap. In the regime we measured, the practical answer to "how should a fixed training budget be spent for calibrated uncertainty?" is: across independent restarts, not along one trajectory.

Declarations

Conflict of interest The author declares that there is no conflict of interest.

AI-use disclosure The experimental code, statistical analyses, figures and initial manuscript draft for this study were produced with the assistance of a large-language-model research workflow (Qinyan Nature Skills) under the author's direction; the author specified the study design, reviewed and edited all content, and verified every reported number against the released per-replicate records. All results are generated by the released code from publicly available datasets.

Data availability The datasets analysed are publicly available: Fashion-MNIST (Xiao et al., 2017), MNIST (LeCun et al., 1998) and KMNIST (Clanuwat et al., 2018). The raw per-replicate metric records, aggregated analysis outputs and per-panel figure source data accompany this paper in the public repository.

Code availability All training, analysis and figure scripts, together with the raw results, are available at `https://github.com/43aquarius/IJMLC` (release `paper-v2.0.0`).

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Supplementary material

Table 4: **Supplementary Table S1 | Pre-specified contrasts on secondary metrics in the core setting** (mean paired difference Δ, 95% CI, exact permutation P; $n = 10$ replicates; uncorrected, descriptive).

Metric	Contrast (A − B)	Δ	95% CI	P_{perm}
Accuracy	DeepEns − Single	+0.0070	[+0.0059, +0.0082]	0.0020
	Snapshot − Single	+0.0040	[+0.0025, +0.0055]	0.0020
	SWA − Single	+0.0031	[+0.0014, +0.0048]	0.0020
	DeepEns − Snapshot	+0.0030	[+0.0019, +0.0041]	0.0039
	Snapshot − SWA	+0.0009	[+0.0004, +0.0013]	0.0078
Brier score	DeepEns − Single	−0.0292	[−0.0311, −0.0274]	0.0020
	Snapshot − Single	−0.0200	[−0.0224, −0.0176]	0.0020
	SWA − Single	−0.0150	[−0.0176, −0.0125]	0.0020
	DeepEns − Snapshot	−0.0093	[−0.0107, −0.0078]	0.0020
	Snapshot − SWA	−0.0050	[−0.0053, −0.0046]	0.0020
ECE (15 bins)	DeepEns − Single	−0.0466	[−0.0481, −0.0450]	0.0020
	Snapshot − Single	−0.0258	[−0.0275, −0.0241]	0.0020
	SWA − Single	−0.0184	[−0.0202, −0.0167]	0.0020
	DeepEns − Snapshot	−0.0208	[−0.0224, −0.0191]	0.0020
	Snapshot − SWA	−0.0073	[−0.0080, −0.0067]	0.0020
AUROC (MSP)	DeepEns − Single	+0.0969	[+0.0847, +0.1091]	0.0020
	Snapshot − Single	+0.0423	[+0.0291, +0.0555]	0.0020
	SWA − Single	−0.0029	[−0.0166, +0.0109]	0.6406
	DeepEns − Snapshot	+0.0546	[+0.0452, +0.0640]	0.0020
	Snapshot − SWA	+0.0452	[+0.0402, +0.0501]	0.0020
AUROC (neg. entropy)	DeepEns − Single	+0.1095	[+0.0970, +0.1220]	0.0020
	Snapshot − Single	+0.0497	[+0.0361, +0.0633]	0.0020
	SWA − Single	+0.0011	[−0.0129, +0.0152]	0.8672
	DeepEns − Snapshot	+0.0598	[+0.0499, +0.0697]	0.0020
	Snapshot − SWA	+0.0485	[+0.0434, +0.0536]	0.0020

Table 5: **Supplementary Table S2 | Exploratory dense weight-averaging variant (*SWA-dense*) in the core setting**, uniform average of all epoch-end checkpoints from epochs 21–60 of the same cyclic trajectories ($n = 10$ replicates; exploratory, uncorrected).

Metric	*SWA-dense* (mean ± s.d.)	vs Single (Δ, 95% CI)	vs Snapshot (Δ, 95% CI)
Test NLL	0.3666 ± 0.0079	−0.1598 [−0.1732, −0.1465]	+0.0182 [+0.0154, +0.0209]
ECE (15 bins)	0.0460 ± 0.0023	−0.0242 [−0.0260, −0.0224]	+0.0016 [+0.0003, +0.0029]
AUROC (MSP)	0.7458 ± 0.0149	−0.0075 [−0.0190, +0.0041]	−0.0498 [−0.0561, −0.0435]

Table 6: **Supplementary Table S3 | Temperature-scaled NLL contrasts versus Single across all settings** (mean paired difference on TS-scaled NLL, 95% CI, Holm-adjusted exact permutation P within each setting's TS family; $n = 10$ replicates each).

Setting	Contrast on TS-NLL	Δ	95% CI	P_{Holm}
Fashion-MNIST, B=60, k=3	DeepEns	-0.0434	$[-0.0468, -0.0400]$	0.0098
	Snapshot	-0.0314	$[-0.0351, -0.0276]$	0.0098
Fashion-MNIST, B=30, k=3	DeepEns	-0.0114	$[-0.0137, -0.0091]$	0.0098
	Snapshot	-0.0140	$[-0.0170, -0.0110]$	0.0098
Fashion-MNIST, B=120, k=3	DeepEns	-0.0567	$[-0.0605, -0.0529]$	0.0098
	Snapshot	-0.0349	$[-0.0375, -0.0323]$	0.0098
Fashion-MNIST, B=60, k=5	DeepEns	-0.0384	$[-0.0423, -0.0345]$	0.0098
	Snapshot	-0.0374	$[-0.0403, -0.0345]$	0.0098
MNIST, B=60, k=3	DeepEns	-0.0071	$[-0.0085, -0.0057]$	0.0098
	Snapshot	-0.0016	$[-0.0032, -0.0001]$	0.0625 (n.s.)

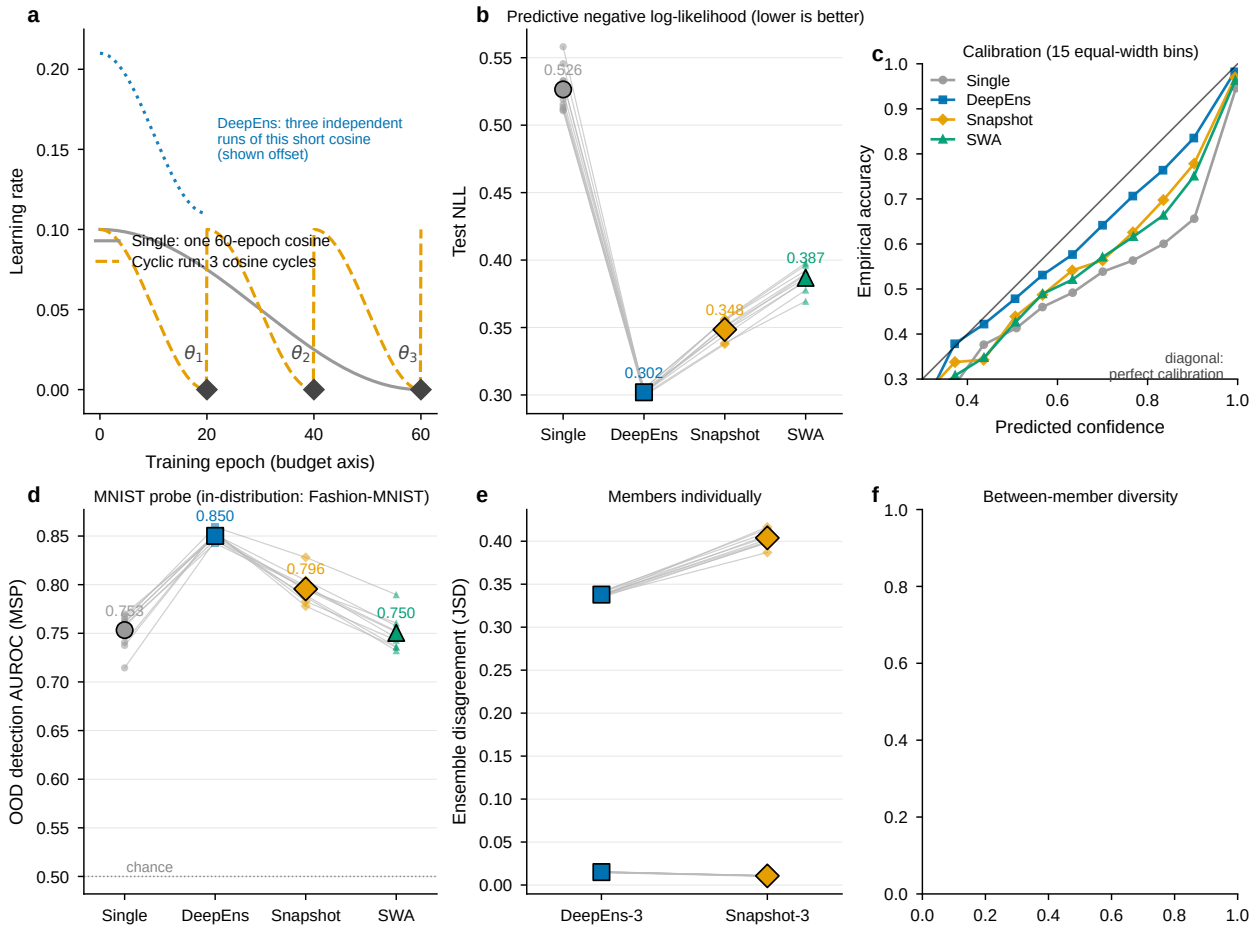


Figure 1: **Budget-matched comparison of four uncertainty strategies (core setting). a**, Learning-rate schedules actually used in training, drawn on a shared budget axis: one 60-epoch cosine anneal (SINGLE); a cyclic schedule of three cosine cycles whose cycle-end checkpoints θ_1–θ_3 (diamonds) form the SNAPSHOT members and the SWA average; and the 20-epoch member schedule of DEEPENS (three independent runs of the schedule shown, offset for display). **b**, Test negative log-likelihood (NLL; lower is better). **c**, Reliability diagrams (15 equal-width confidence bins, mean over replicates); the diagonal marks perfect calibration. **d**, OOD-detection AUROC using the maximum-softmax score on an MNIST probe (in-distribution: Fashion-MNIST test); the dotted line marks chance. **e**, Mean test NLL of individual ensemble members. **f**, Between-member disagreement (mean pairwise Jensen–Shannon divergence, JSD). In **b**, **d**–**f**, thin grey lines connect the ten replicate-level observations (small markers; conditions additionally distinguished by marker shape as well as colour), large outlined markers show condition means ($n = 10$ replicates; all observations shown; no exclusions). Ensemble size $k = 3$ throughout.

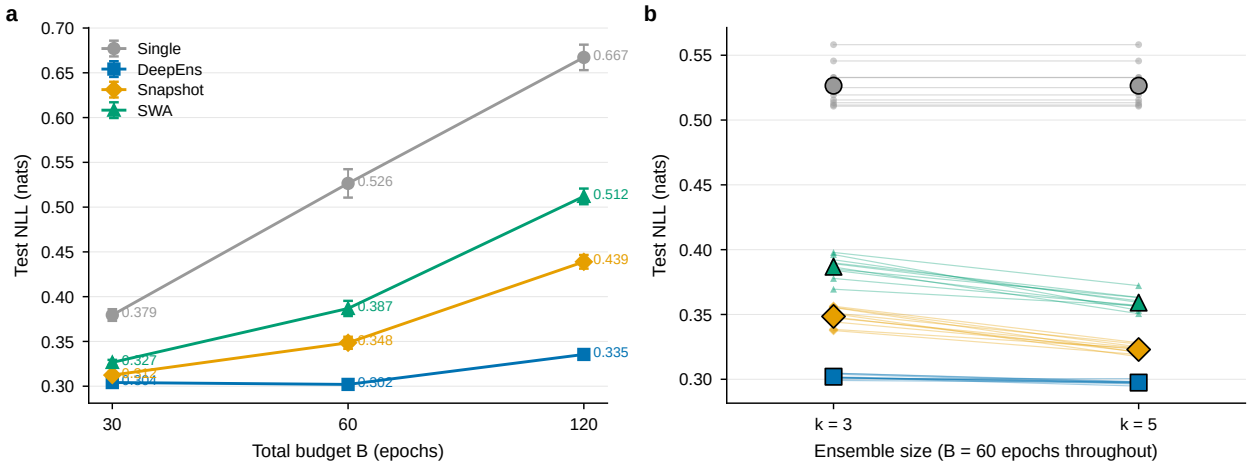


Figure 2: **Budget and ensemble-size robustness (Fashion-MNIST). a**, Test NLL by condition as the total budget varies ($B = 30$, 60, 120 epochs; $k = 3$; mean ± s.d., $n = 10$ replicates). All strategy-versus-SINGLE contrasts are significant at each budget (Holm-adjusted $P = 0.0098$); the budget-split advantage widens as longer training makes the single network more overconfident. **b**, Test NLL at $k = 3$ versus $k = 5$ ($B = 60$; thin lines connect the ten replicate-level observations; large markers are condition means). Enlarging the ensemble improves every strategy and preserves the ordering.

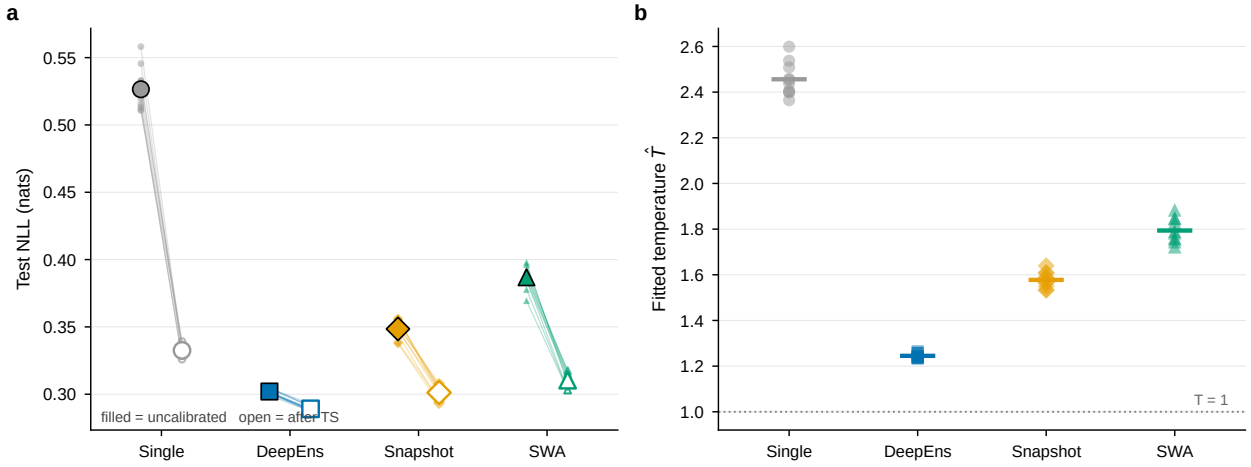


Figure 3: **Per-condition temperature scaling (core setting). a**, Test NLL before (filled) and after (open) fitting one temperature per condition and replicate on validation data; thin lines connect the ten replicate-level observations, large markers are condition means. TS repairs the single network more than any ensemble, yet the budget-split ensemble remains significantly best (ΔNLL -0.0434 after TS, Holm-adjusted $P = 0.0098$). **b**, Fitted temperatures $\hat{T}$ (replicate dots; horizontal bars are means; the dotted line marks $T = 1$, i.e. no scaling). The ordering of $\hat{T}$ mirrors the uncalibrated overconfidence ordering.

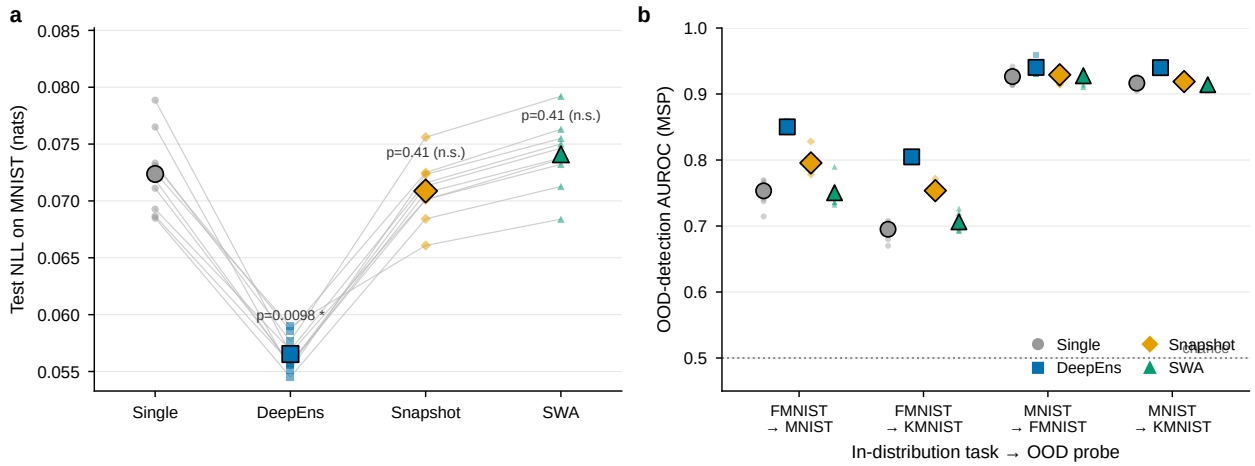


Figure 4: **Reversed dataset direction and OOD overview. a**, Test NLL on MNIST as the in-distribution task ($B = 60$, $k = 3$; thin lines connect replicate-level observations, large markers are condition means). Only the budget-split ensemble separates from the single network (Holm-adjusted $P = 0.0098$; the single-trajectory strategies are not significant, $P = 0.41$). **b**, OOD-detection AUROC (MSP score) for the four conditions across all four in-distribution→probe combinations (small dots: ten replicates; large markers: condition means). DeepEns is best on every combination; SWA tracks Single throughout. The dotted line marks chance.

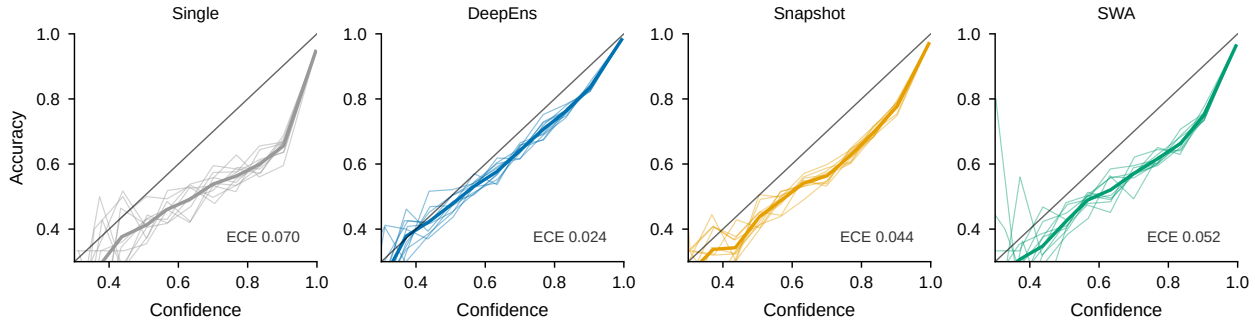


Figure 5: **Supplementary Figure S1 | Reliability diagrams per condition (core setting).** Thin lines show the ten individual replicates; the bold line shows the replicate-mean reliability curve (15 equal-width bins). ECE (mean over replicates) is annotated in each panel. The single network's curve lies furthest below the diagonal at high confidence; DeepEns lies closest to it.

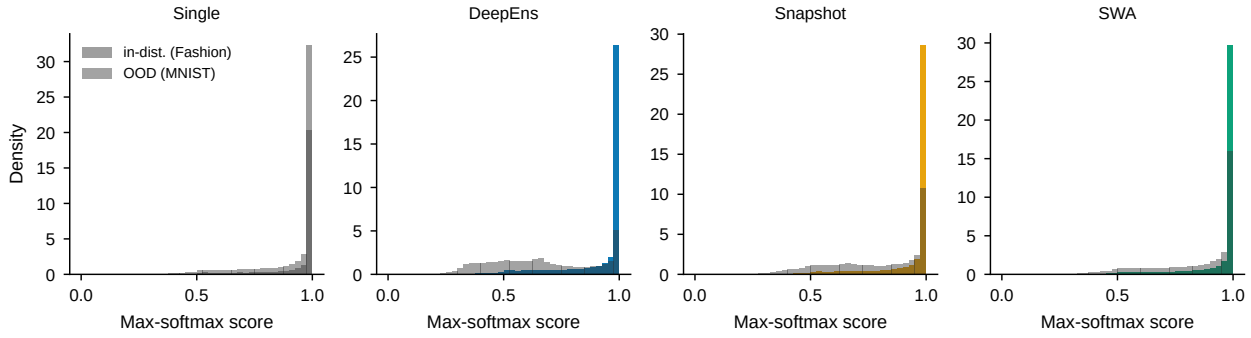


Figure 6: **Supplementary Figure S2 | Maximum-softmax score distributions underlying the OOD analysis (core setting).** Mean score density over the ten replicates, in-distribution (Fashion-MNIST test; coloured) versus OOD probe (MNIST; grey). The two prediction-averaging strategies shift OOD scores leftward (lower confidence on shifted inputs) substantially more than SWA does, which is the direct visual counterpart of the AUROC result in Fig. 1d.

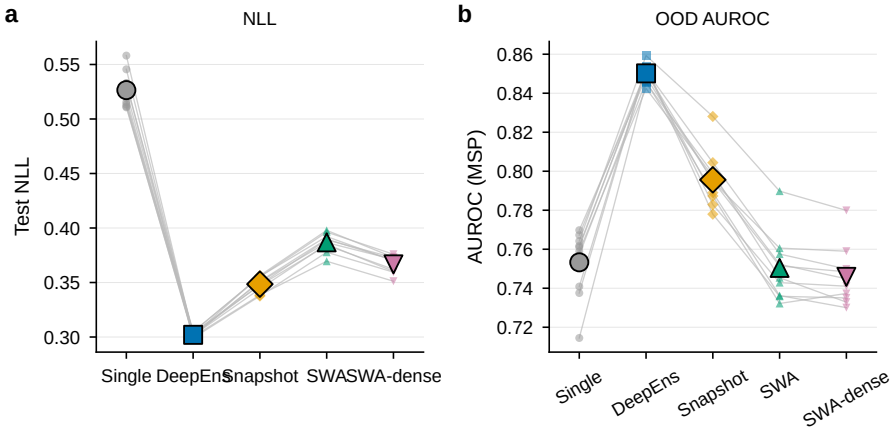


Figure 7: **Supplementary Figure S3 | Exploratory dense weight-averaging variant (core setting).** **a**, Test NLL and **b**, OOD AUROC for the pre-registered conditions plus the exploratory *SWA-dense* variant (uniform average of all epoch-end checkpoints from epochs 21–60 of the same cyclic trajectories). Dense averaging improves the in-distribution metrics over SWA but does not close the gap to the snapshot ensemble and leaves OOD detection unchanged from Single.

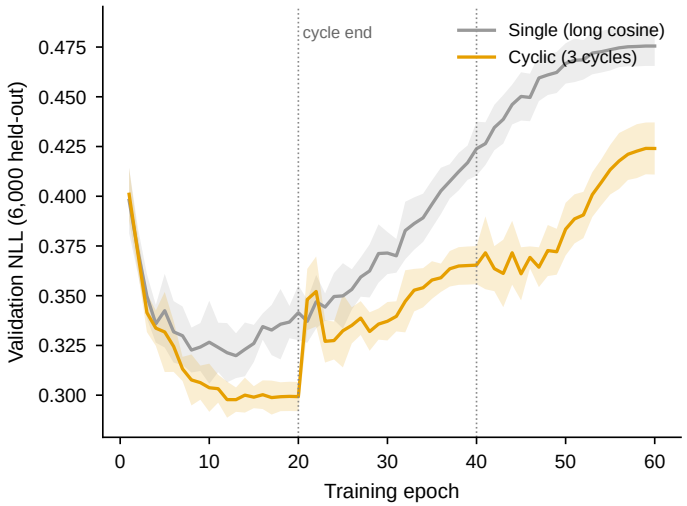


Figure 8: **Supplementary Figure S4 | Validation learning curves of the two training trajectories (core setting).** Mean ± s.d. of validation NLL (6,000 held-out images) over the ten replicates for the single 60-epoch cosine anneal and the cyclic three-cycle run. Dotted vertical lines mark cycle boundaries (epochs 20 and 40).